EXAM Linear Control Design 1

25 May 2022 Questions - Multiple Choice

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1. Q1. Consider the block diagram shown in the following figure Fig.1. **What is the transfer function** $\frac{Y(s)}{R(s)}$ (Select the correct answer)? (1 Point)

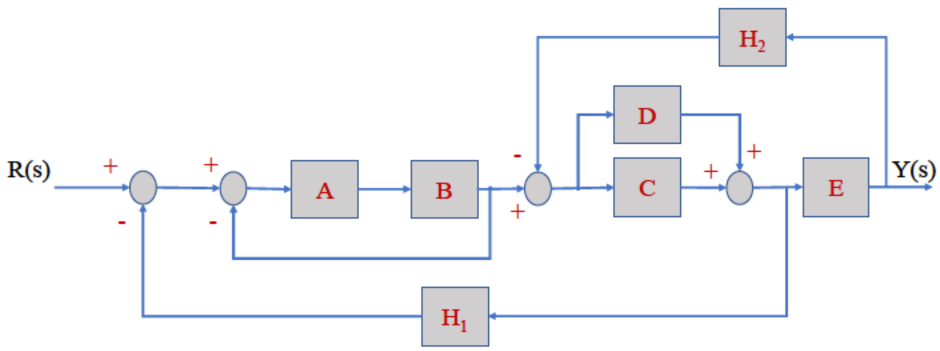


Figure 1: Block Diagram

1. $$\frac{Y(s)}{R(s)} = \frac{ABE^2(C+D)}{(1+AB)[1+(C+D)EH_2]E + ABE(C+D)H_1}$$

   [redacted]

2. $$\frac{Y(s)}{R(s)} = \frac{AB(C+D)}{(1+AB)[E+(C+D)H_2] + AB(C+D)H_1}$$

   [redacted]

3. $$\frac{Y(S)}{R(s)} = \frac{ABE^2(C+D)}{(1+AB)[1+(C+D)H_2]E - AB(C+D)H_1}$$

   [redacted]

4. $$\frac{Y(s)}{R(s)} = \frac{AB(C+D)}{(1+AB)[E+(C+D)H_2]E - AB(C+D)H_1}$$

   [redacted]

[redacted]

2. Q2. Consider the following transfer function of the RC circuit: $G(s) = \frac{V_c(s)}{V_i(s)} = \frac{\frac{1}{RC}}{s+\frac{1}{RC}}$, where V_c is the voltage across the capacitor, V_i is the source voltage, R is the resistance (50Ω), and C is the capacitance (160F). **Which of the following statements is FALSE?**

   1. The time duration after which the capacitor voltage has reached the steady state value is 40ms

      [redacted]

   2. The time constant TAU is the time taken for the capacitor voltage to reach 63.2% of its final value from t=0

      [redacted]

   3. The initial value of the voltage V_c is 0

      [redacted]

   4. The final value of the unit step response is 1

      [redacted]

   5. The time duration after which the capacitor voltage has reached the 63.2% of its final steady state value is 16 ms

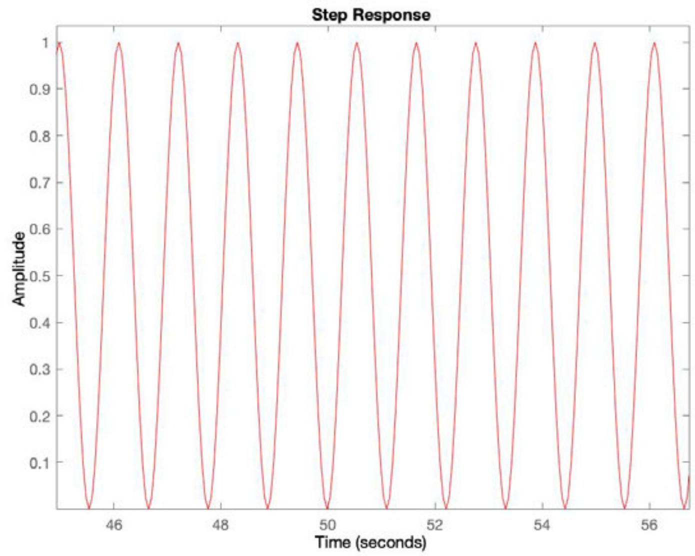
      [redacted]

[redacted]

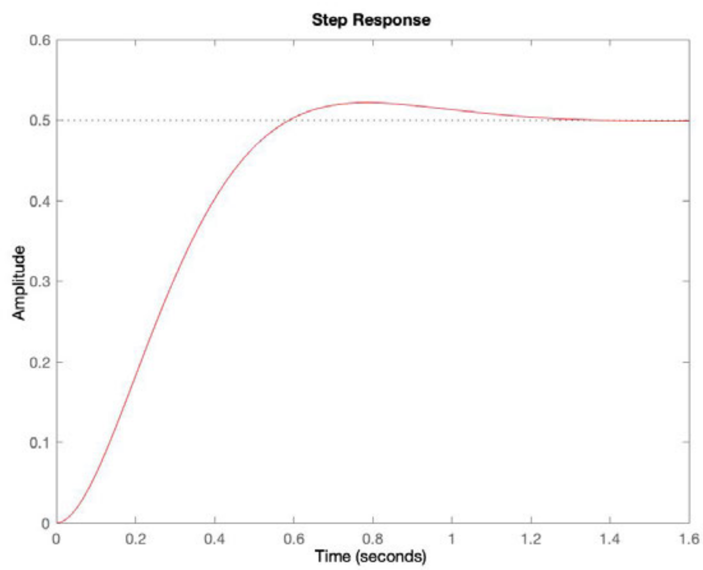
3. Q3. Consider a second order system

   $$G(s) = \frac{w_n}{s^2 + 2\zeta w_n s + w_n^2}$$

Select the correct unit step response when $\zeta = 0$

1. 

[redacted]

2. 

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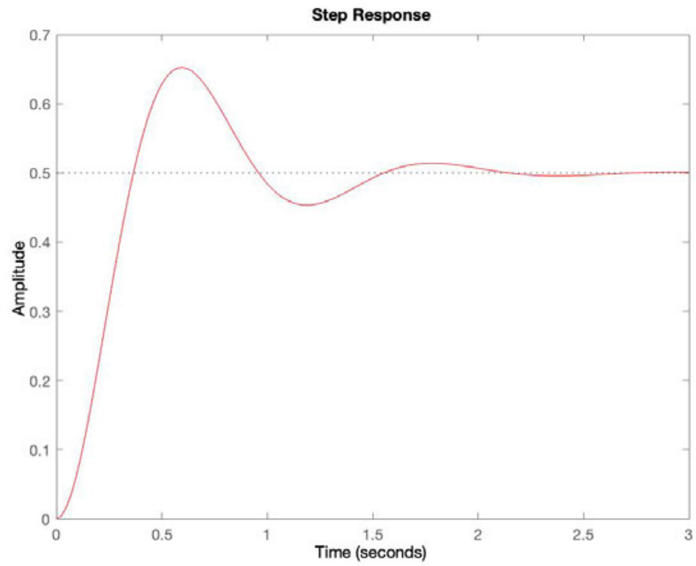
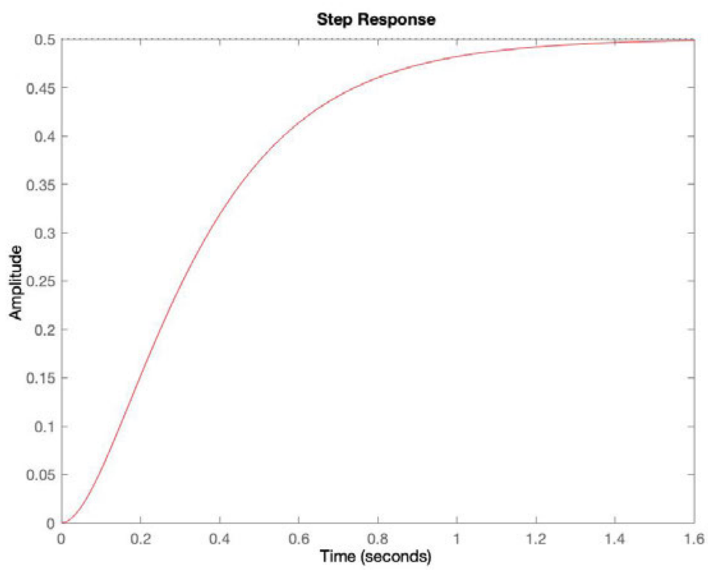


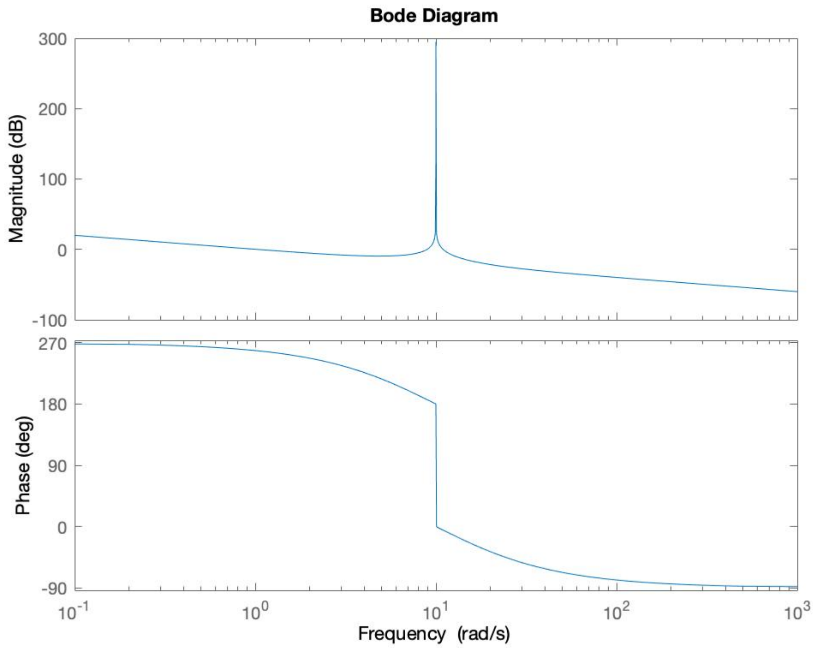
3. [redacted]



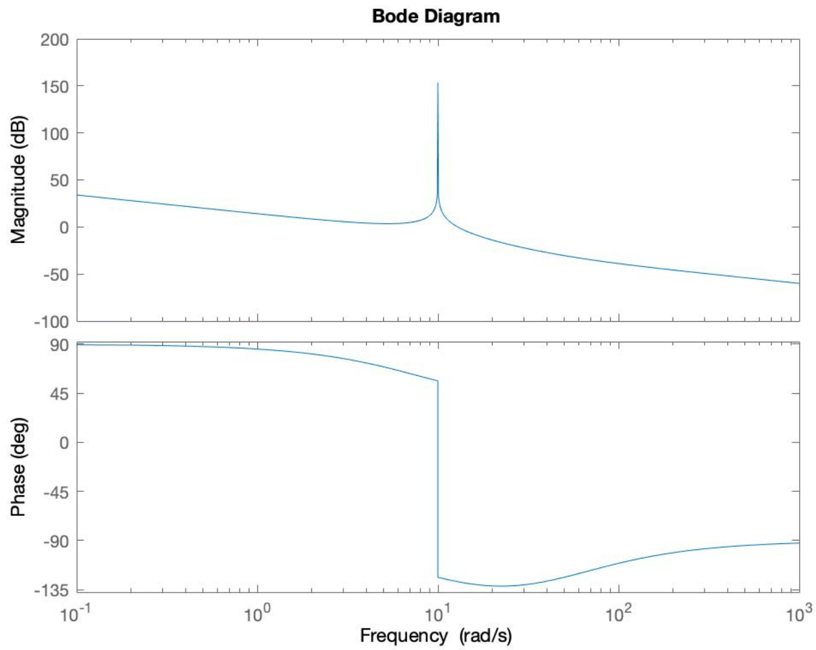
4. [redacted]

[redacted]

4. Q4. An open-loop transfer function has one real zero in the RHP and two conjugate complex poles (null real part):
Which is the corresponding Bode plot? (1 Point)



1.
[redacted]



2.

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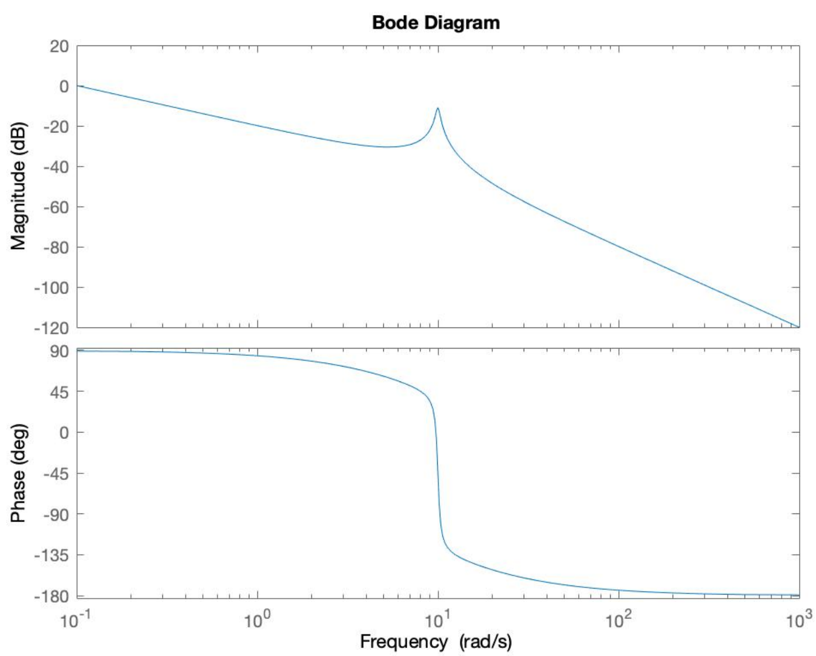
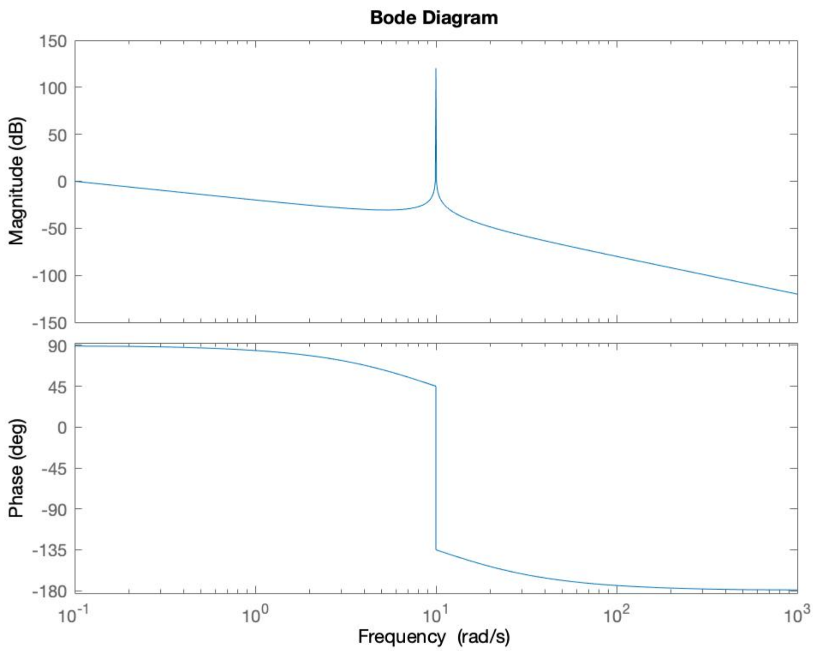


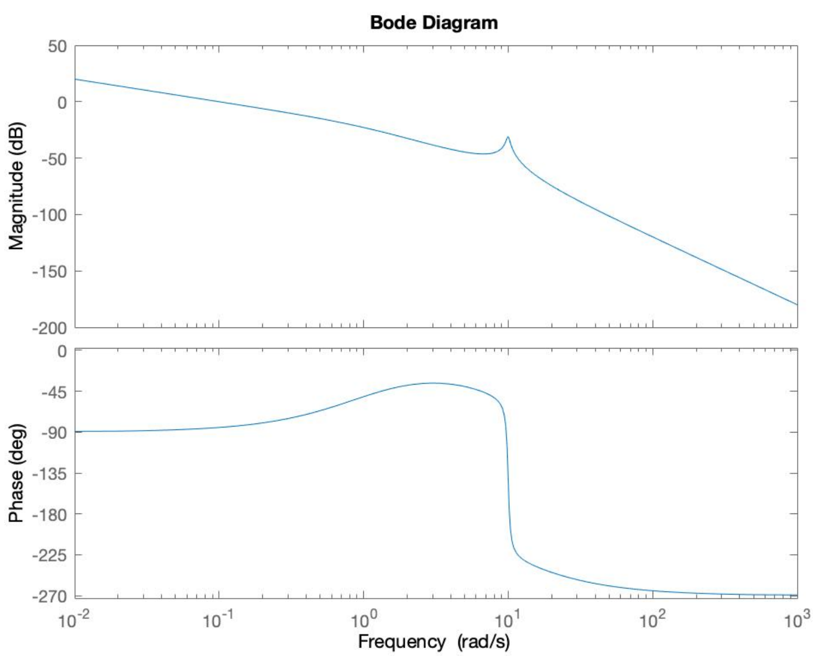
[redacted]

[redacted]



4.

[redacted]



[redacted] (pole in

[redacted]

[redacted]

5. Q5. The Bode plot diagrams of the magnitude and phase of a transfer function G(s) of a linear system is shown in Fig. 2.
Which of the following answers is correct? (1 Point)

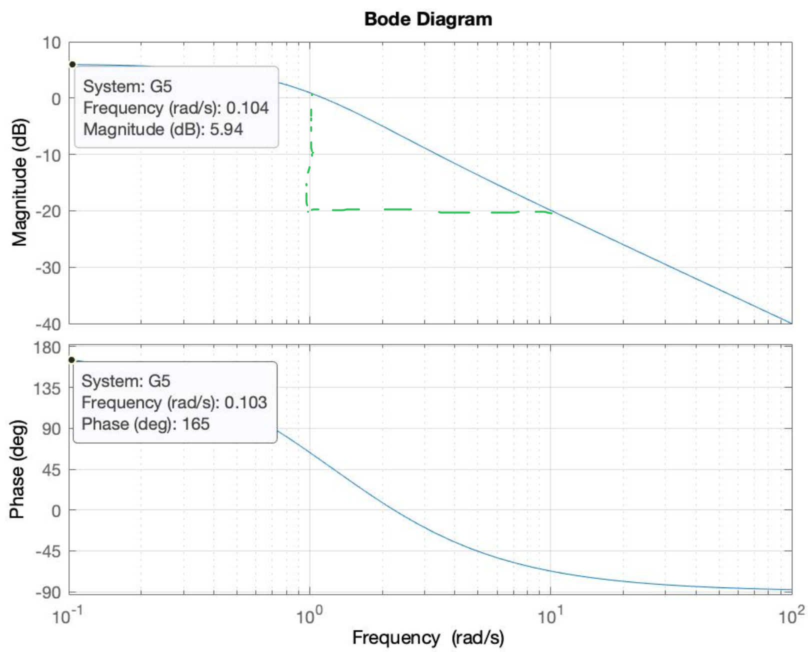


Figure 2: Bode Plot Diagrams - Q5

1. G(s) has negative and positive real poles and no zeros
[redacted]
2. G(s) has only 1 negative real pole and 1 negative real zero
[redacted]
3. G(s) has only 1 positive real pole and no zeros
[redacted]
4. G(s) has complex poles and 1 negative real zero
[redacted]

5. G(s) has positive real poles and 1 positive real zero
   [redacted]

[redacted]

[redacted] $\frac{[redacted]}{[redacted]}$

6. Q6. A closed-loop system has a loop transfer function

$$G(s) = \frac{K}{s(s+21)}$$

and the following Bode plot in Fig.3

What is the gain K so that the phase margin is PM=40°?
(1 Point)

1. K = 0.1 [redacted]
2. K = 8.4 [redacted]
3. K = 77.5 [redacted]
4. K = 18.5 [redacted]
5. K = 40 [redacted]

[redacted]

7. Q7. **What is the DC gain in Db of the following transfer function?**

$$G(s) = \frac{12}{(s+2)(s+3)}$$

(1 Point)

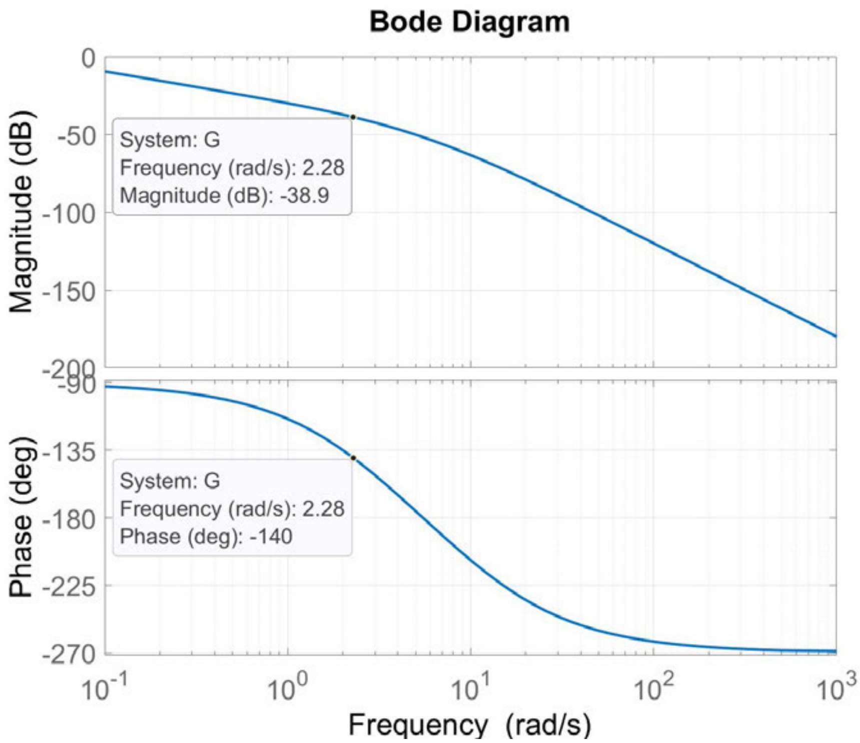


Figure 3: Bode Plot - Q6

1. DC = 2 [redacted]
2. DC = 12 [redacted]
3. DC = 6 [redacted]
4. DC = 0 [redacted]
5. DC = -15.6 [redacted]

[redacted]

8. Q8. Consider the differential equation

   $5\ddot{y}(t) + \dot{y}(t) + 0.5y(t) = 3u(t)$

   Where y(0) = $\dot{y}(0)$ = 0 and u(t) is the input.
   Which are the poles of this system? (1 Point)

1. s1 = -0.1, s2= -0.1 [redacted]
2. s1 = -0.1+0.3j, s2= -0.1-0.3j [redacted]
3. s1 = 0.3j, s2= -0.3j [redacted]
4. s1 = 0.3, s2= 0.1 [redacted]
5. s1 = 0.1+0.3j, s2= 0.1-0.3j [redacted]

[redacted]

9. Q9. Consider the following Linear Time Varying system:

$$\dot{x}_1 = -x_1 + x_2$$

$$\dot{x}_2 = 2x_1 - wx_2$$

For which values of the parameter w are the poles of the system in the left half-plane? (1 Point)

1. $w \leq 2$
   [redacted]
2. $w \geq 1$
   [redacted]
3. $w < 1$
   [redacted]
4. $w > 2$
   [redacted]
5. $w \geq -2$
   [redacted]

[redacted]

10. Q10. A system with the following characteristic equation is $s^2 + 2s + 2 = 0$ of the closed loop system is called (select the right answer): (1 Point)

    1. Over damped

       [redacted]

    2. Undamped

       [redacted]

    3. Underdamped

       [redacted]

    4. Critically damped

       [redacted]

[redacted]

Linear Control Design 1
Exam Questionnaire

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Question 11

Consider a third order, open-loop stable system described by a transfer function $G(s)$ with Nyquist diagram as shown in the following figure. The Nyquist curve intersects the negative

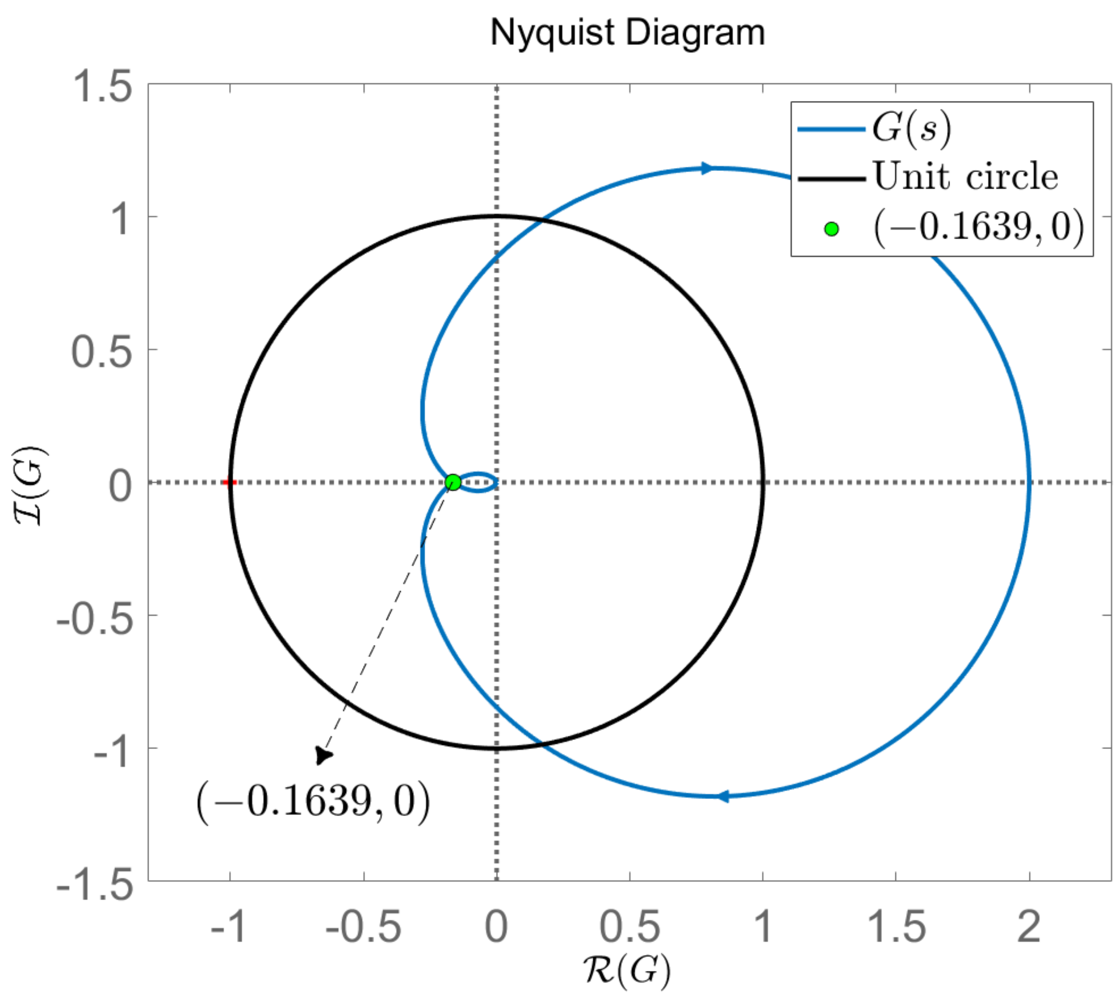


Figure 11: Nyquist diagram

horizontal semi-axis at the point $(-0.1639, 0)$ along the clockwise direction. The gain margin K_M in dB is approximately:

a) $K_M = -15.71$

b) $K_M = 15.71$

c) $K_M = -6.1$

d) $K_M = 6.1$

e) $K_M = 0.1639$

Question 12

Consider a system described by a transfer function $G(s)$ with poles $\{4.5,\ -3 \pm j\}$ and open-loop Nyquist plot as shown in the following figure. The Nyquist plot intersects the real axis

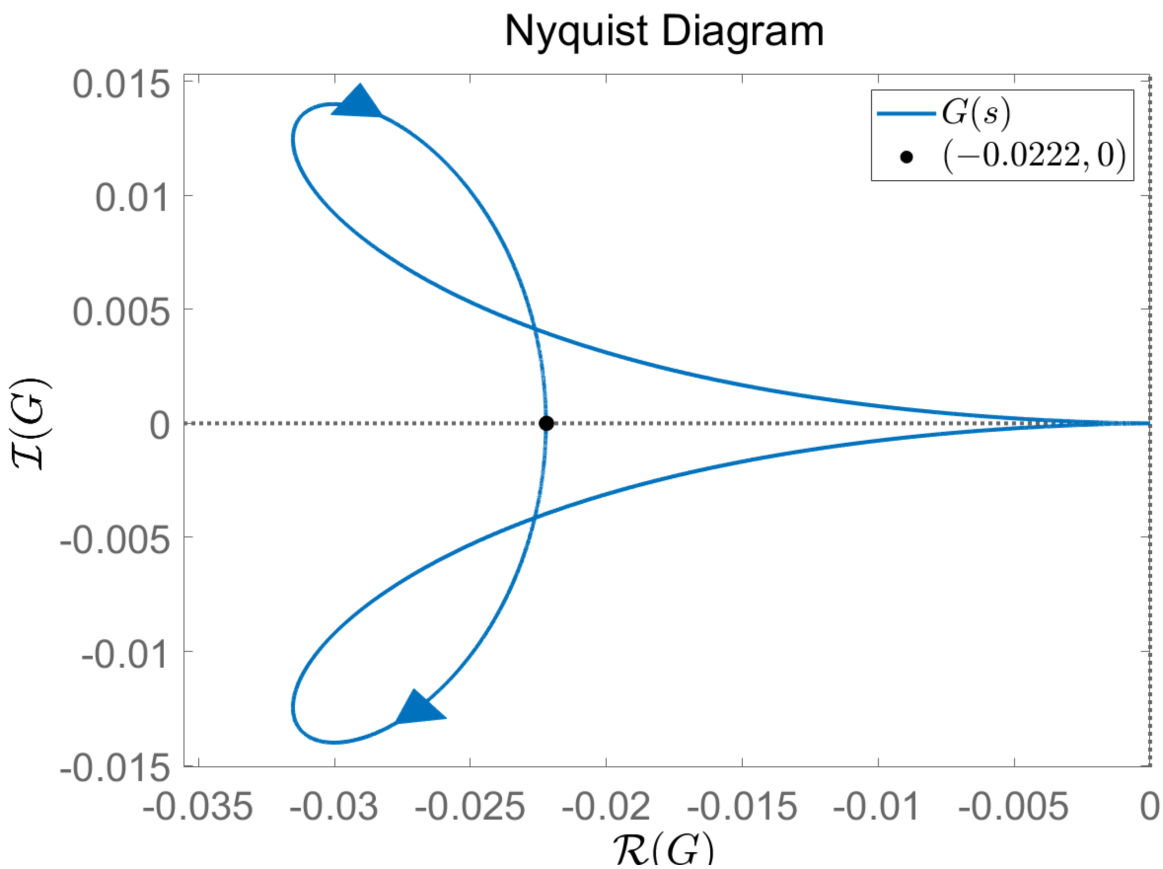


Figure 12: Nyquist plot

along the counter-clockwise direction at the point $(-0.0222, 0)$. A P-controller K_P is applied to the system and the loop is closed with unit feedback. The closed-loop system is stable for:

a) $0.0222 < K_P < 1$

b) $K_P = 50$

c) $-45 < K_P < 0$

d) $-0.0222 < K_P < 0$

e) $K_P = 0.0222$

Question 13

Consider an open-loop system $G_{ol}(s) = K_P C_{PI}(s) C_D(s) G(s)$, where K_P is the proportional gain, $G(s)$ is the transfer function of the system to be controlled, C_{PI} is the PI controller part and the Lead part $C_D(s)$ is defined as

$$C_D(s) = \frac{0.355s + 1}{\alpha 0.355s + 1} \, .$$

If the magnitude curve in the Bode plot of $G_{ol}(s)$ is strictly decreasing and $|G_{ol}(10j)| = 1$, what is (approximately) the magnitude contribution M_D in dB of the Lead part to the open-loop transfer function at the crossover frequency?

a) $M_D = 5.5$ dB

b) $M_D = 3.55$ dB

c) $M_D = 11$ dB

d) $M_D = 1.08$ dB

e) $M_D = 0.7$ dB

Question 14

The Bode plot of a third-order stable closed-loop system is shown below: Which one of the following figures corresponds to the error response to a unit step change in the reference signal?

a) (Non zero steady state error)

Question 15

Consider a fourth-order type-0 system with real poles and a real zero. All poles and the zero are located on the left half complex plane, with the zero located at a lower frequency compared to any of the poles. A P-controller with $K_P > 0$ is applied to the system. Which of the following statements is the correct one?

a) The unit-feedback closed-loop system is stable for any $K_P > 0$.

b) Eliminating the steady-state error due to a step change in the reference requires only choosing an appropriate value for K_P.

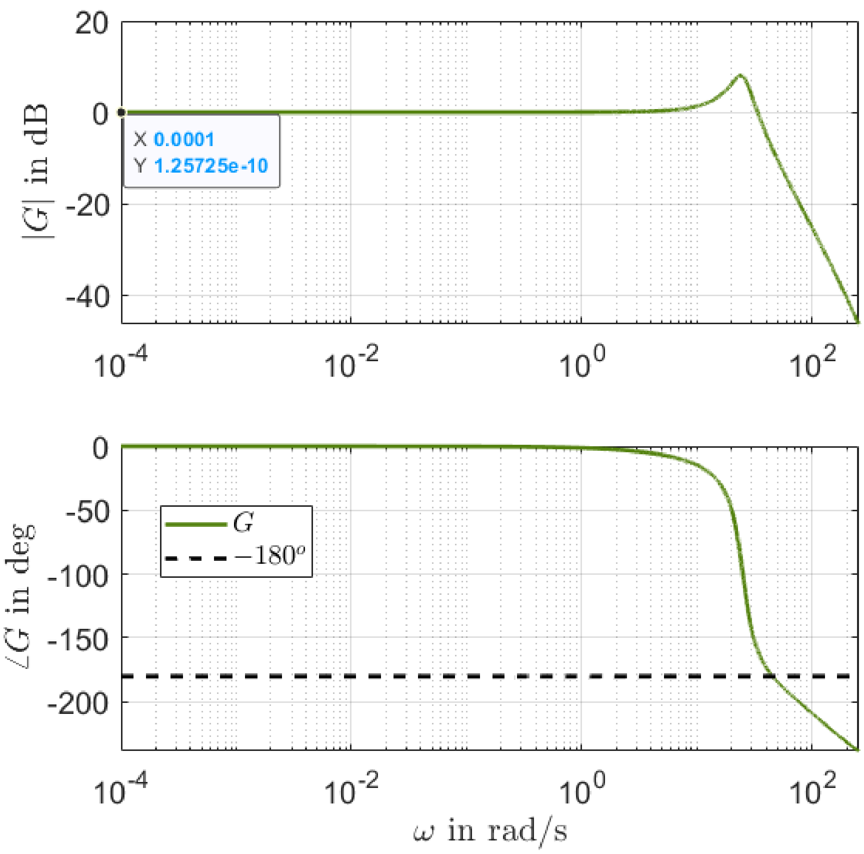


Figure 13: Bode plot

b)

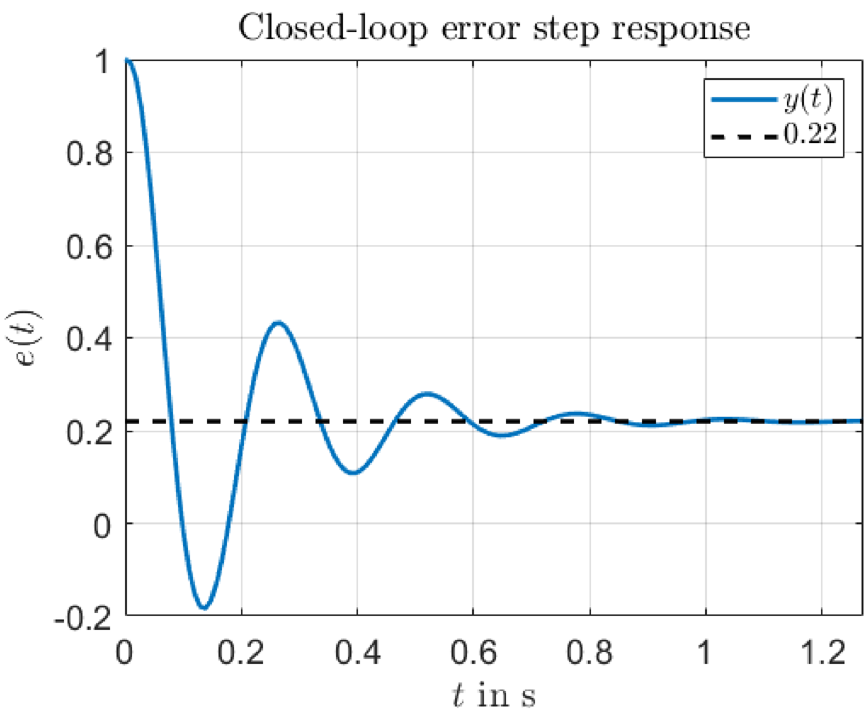
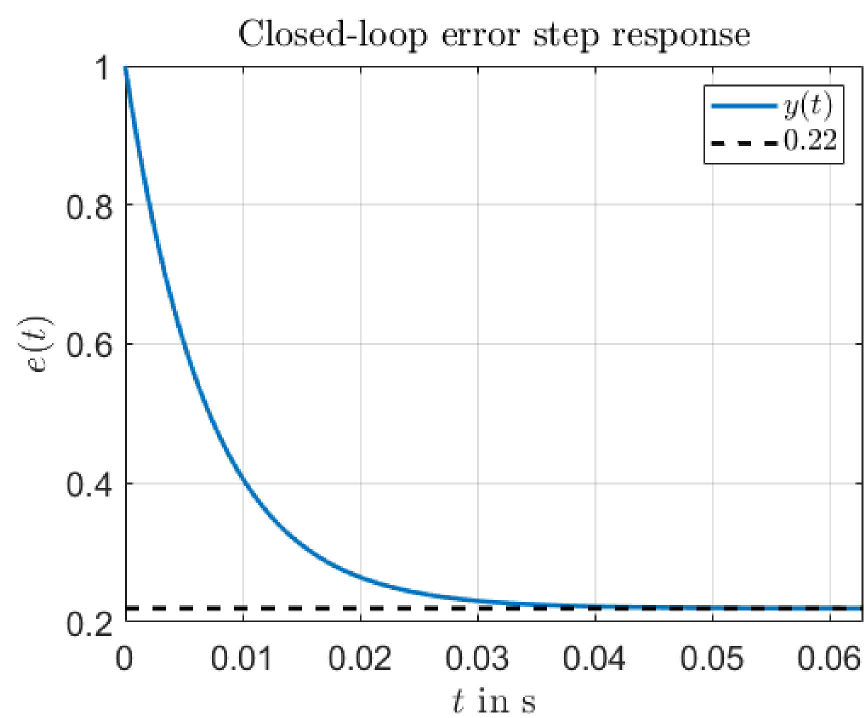
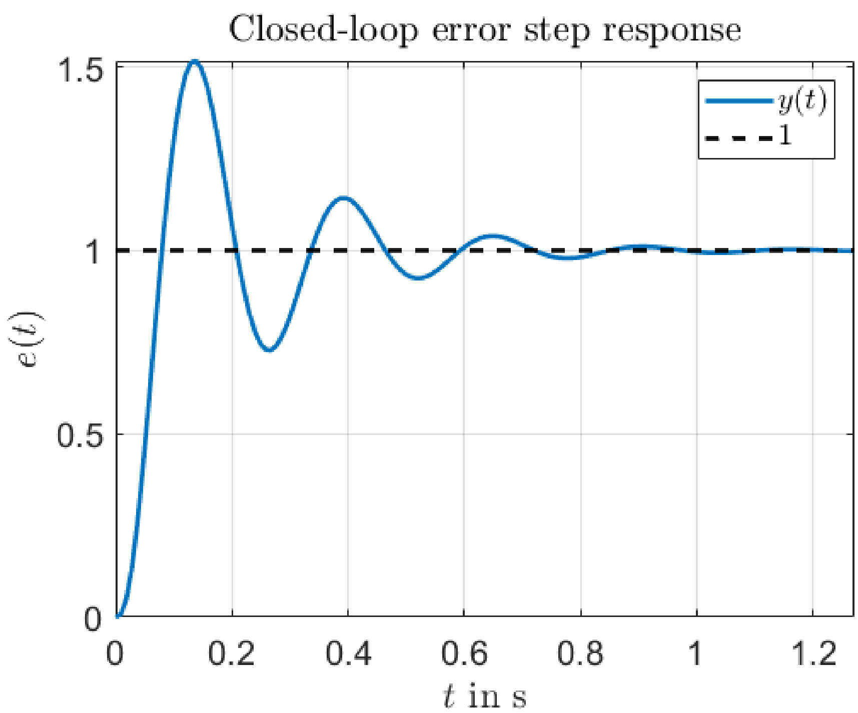


Figure 14: (a) Error step response



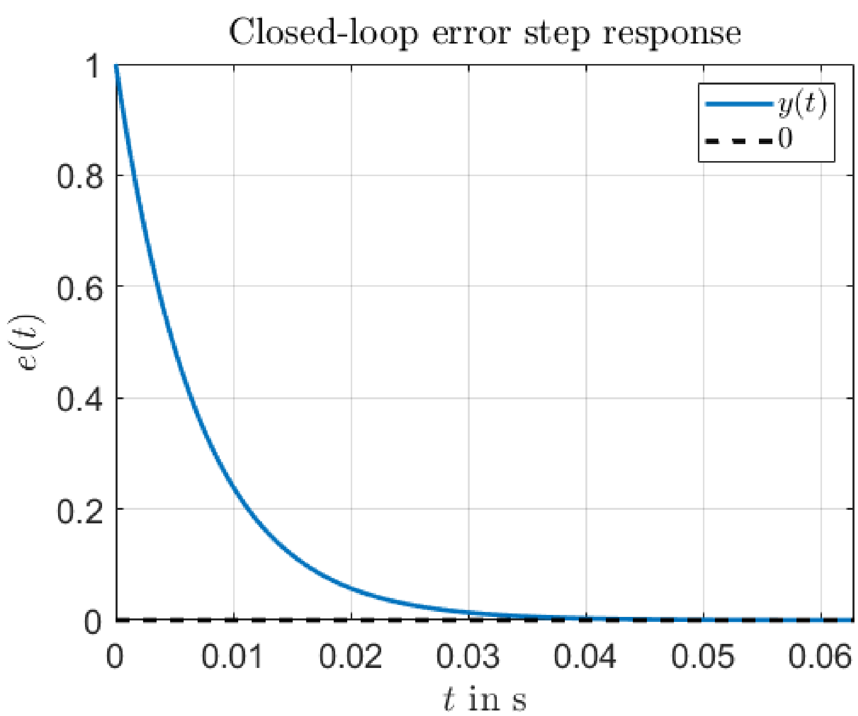
b)

Figure 15: (b) Error step response



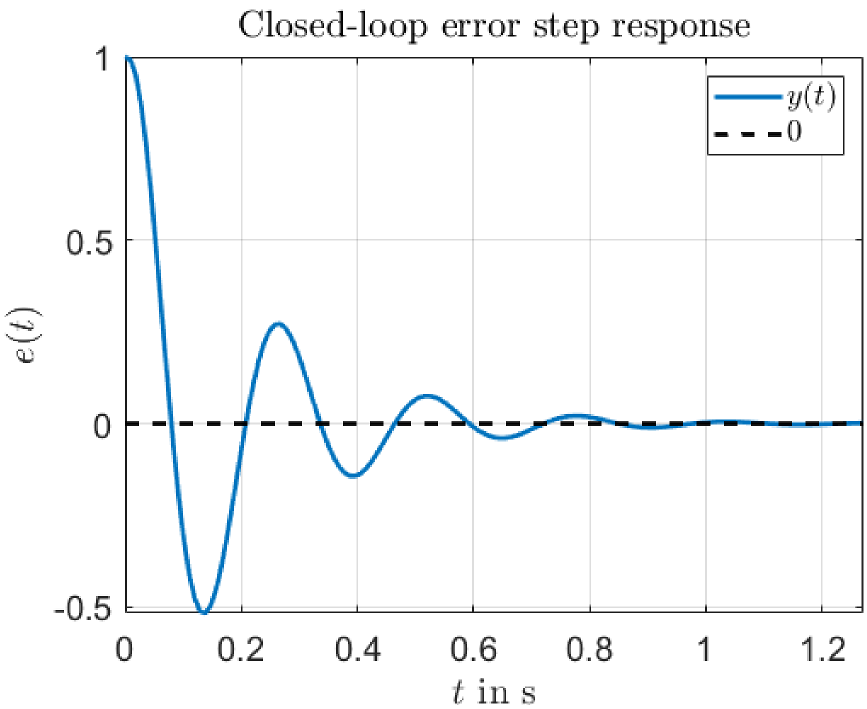
c)

Figure 16: (c) Error step response



d)

Figure 17: (d) Error step response



e)

Figure 18: (e) Error step response

c) There is no positive value of K_P that can give a negative phase margin for the open-loop system.

d) For some values of K_P it is possible to have two crossover frequencies.

e) Stability of the closed-loop system is ensured if K_P makes the Nyquist plot of the open-loop system encircle the point $(-1, 0)$ once along the counter-clockwise direction.

Question 16

Consider a system described by a transfer function $G(s)$ with a magnitude Bode plot as shown in the figure below: A P-controller with gain K_P is applied to the system and the loop is closed with unit feedback as in the following figure. If the steady state error $e_{SS} = \lim_{s \to 0} e(s)$ due to a unit step change in the reference signal is $e_{SS} = 0.555$, the proportional gain K_P is approximately:

a) $K_P = 1$

b) $K_P = 2$

c) $K_P = 2.5$

d) $K_P = 0.4$

e) $K_P = 7.9588$

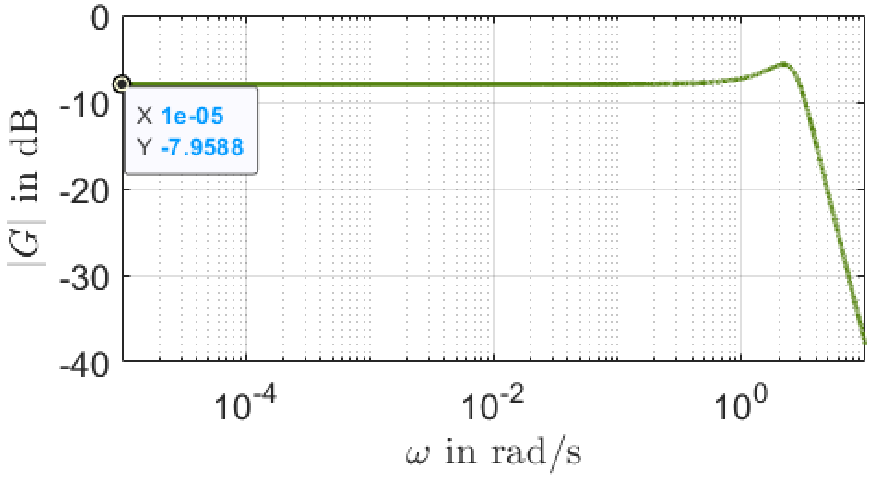


Figure 19: Bode plot.

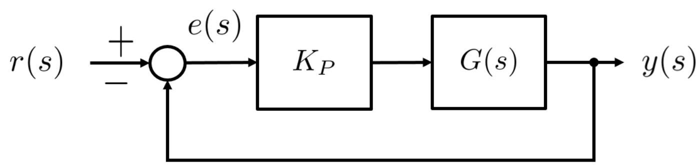


Figure 20: Closed-loop system.

Question 17

Consider the system described by a transfer function G with a phase Bode plot as shown in the following figure. The objective is to design a PI-Lead controller $C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s}$ such that the open-loop system has crossover frequency $\omega_c = 6.4$ rad/s and a phase margin $\gamma_M = 75^o$ is achieved. If the zero of the PI part is placed at a frequency 5 times smaller than the crossover frequency, the α parameter of the Lead part is approximately:

a) $\alpha = 0.032$

b) $\alpha = 2$

c) $\alpha = -1.5$

d) $\alpha = 1.13$

e) $\alpha = 0.5$

Question 18

Consider the system described by the following block diagram where $n > 2$ is a positive integer

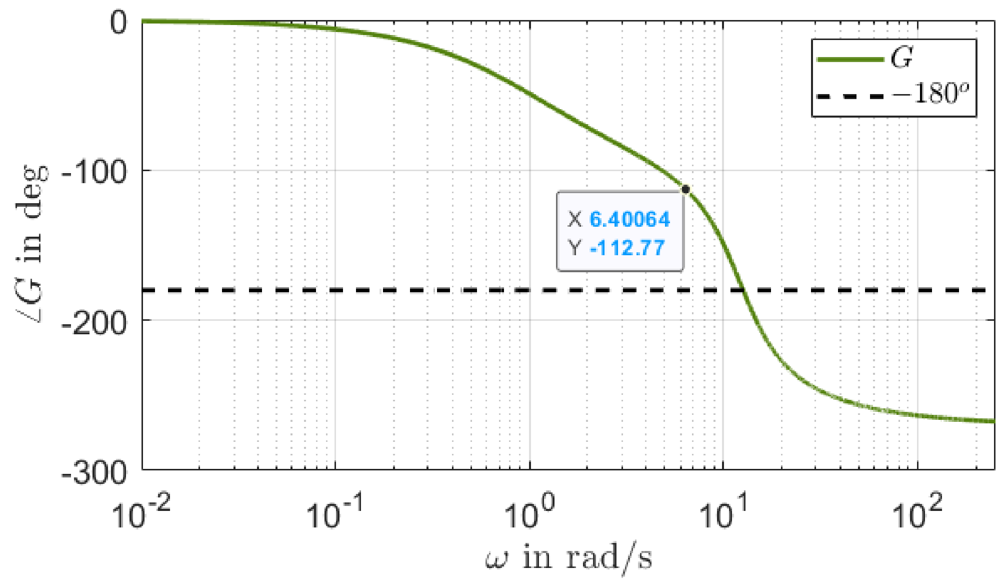


Figure 21: Bode plot

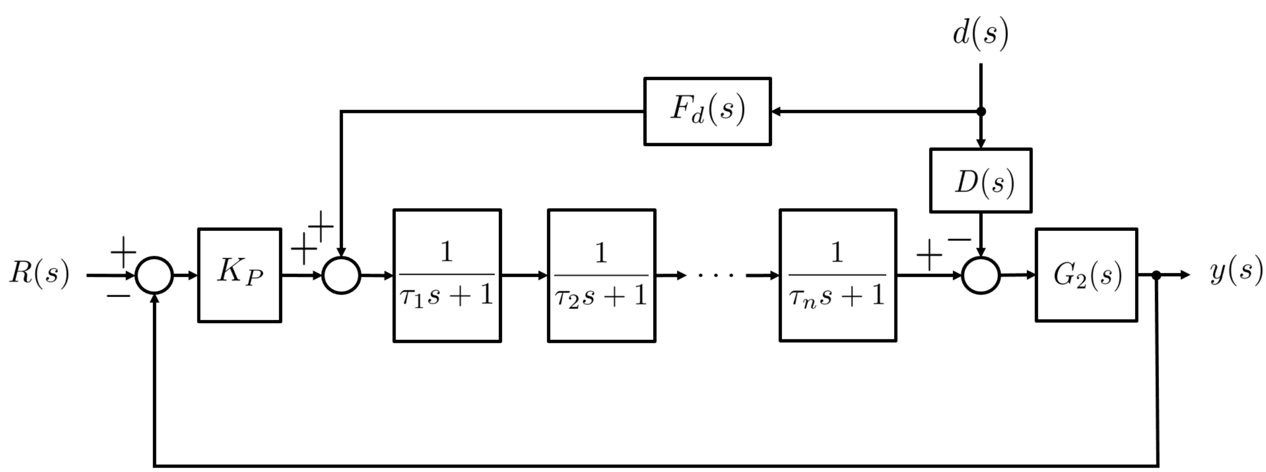


Figure 22: Block diagram.

number and the time constants $\tau_k > 0,\ k = 1, \ldots, n$ are known. The objective is to attenuate the effect of the disturbance $d(s)$ on the system output $y(s)$ by designing a disturbance feed-forward controller $F_d(s)$. If the known transfer function $D(s)$ is defined as

$$D(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \text{ with } \omega_n > 0 \text{ and } 0 < \zeta < 1\ ,$$

which one of the following designs for $F_d(s)$ is the most appropriate?

Hint: *The product operator* $\prod_{k=1}^{n}$ *is defined as* $\prod_{k=1}^{n} A_k = A_1 \cdot A_2 \cdot A_3 \cdot \ldots \cdot A_n$.

a) $F_d(s) = \dfrac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{s^2 + 2\zeta\omega_n s + 1}$

b) $F_d(s) = \dfrac{(s^2 + 2\zeta\omega_n s + 1) \prod_{k=1}^{n} (\tau_k s + 1)}{\omega_n^2 (\tau_f s + 1)^{n+2}}$ with $\tau_f \leq \dfrac{\min(\tau_1, \ldots, \tau_n)}{5}$

c) $F_d(s) = \dfrac{\prod_{k=1}^{n} (\tau_k s + 1)}{(\tau_f s + 1)^n}$ with $\tau_f \leq \dfrac{\min(\tau_1, \ldots, \tau_n)}{5}$

d) $F_d(s) = \dfrac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{(s^2 + 2\zeta\omega_n s + 1)(\tau_f s + 1)^{n-2}}$ with $\tau_f \leq \dfrac{\min(\tau_1, \ldots, \tau_n)}{5}$

e) $F_d(s) = \dfrac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{(s^2 + 2\zeta\omega_n s + 1)(\tau_f s + 1)^n}$ with $\tau_f \geq \dfrac{\max(\tau_1, \ldots, \tau_n)}{5}$

Question 19

A system is described by the following transfer function

$$G(s) = \frac{900}{(0.25s + 1)(s^2 + 50s + 3000)}$$

A PI-Lead controller

$$C(s) = K_P \underbrace{\frac{\tau_i s + 1}{\tau_i s}}_{C_{PI}(s)} \underbrace{\frac{\tau_d s + 1}{\alpha \tau_d s + 1}}_{C_D(s)}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 75^o$. If ω_c is the open-loop crossover frequency, $N_i = \omega_c \tau_i = 3$ and $\alpha = 0.001$, the controller proportional gain K_P is approximately:

a) $K_P = 3.4154$.

b) $K_P = 10.67$.

c) $K_P = 36$.

d) $K_P = 0.293$.

e) $K_P = -10.67$.

Question 20

Which of the following statements is correct?

a) Higher phase margin of the open-loop transfer function implies reduced oscillations in the closed-loop step response and more robustness.

b) Increasing the parameter α in a PI-Lead controller design improves the phase margin of the open loop system.

c) Lower bandwidth of a closed-loop system implies faster response of the output to changes in the reference signal.

d) Moving the Lead part of a PI-Lead controller from the feedback branch to the forward branch decreases the effect of an input disturbance (i.e. a disturbance that is added to the input signal) on the step response of the closed-loop system.

e) Moving the Lead part of a PI-Lead controller from the feedback branch to the forward branch improves oscillation attenuation for the step response of the closed-loop system.